PROPOSAL

"Edge AI-Based Industrial Sorting System using YOLOv8 on Raspberry Pi 5 with Omron PLC Integration"

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PROPOSAL

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We have prepared this Research Internship Submission Proposal with the hope of conducting a research internship at the Faculty of Computer Science, Universitas Pembangunan Nasional Veteran Jawa Timur.

BAB I

INTRODUCTION

1.1. Research Title : Edge AI-Based Industrial Sorting System using YOLOv8 on Raspberry Pi 5 with Omron PLC Integration

1.2. Problem Formulation

Traditional industrial sorting systems often require expensive and complex specialized hardware for integration. Although *edge computing* devices offer a more affordable solution, the standard CPU on these devices often experiences an **inference bottleneck** when running *deep learning* models like YOLOv8 in *real-time*. The main challenge in this research is how to optimize the **Raspberry Pi 5** with the support of the **Hailo-8 AI Accelerator (26 TOPS)** to achieve inference speed (high FPS) and accuracy equivalent to industry standards. Furthermore, an efficient communication mechanism is required between the AI perception layer (Raspberry Pi) and the industrial execution layer (**Omron PLC**) to ensure a fast and synchronized system response.

1.3. Activity Objectives

The objectives of this activity are to:

1. **Optimization and Implementation:** Conduct training and compilation of the YOLOv8 model into the **Hailo Executable Format (.hef)** to maximize *edge computing* performance on the Raspberry Pi 5.
2. **System Integration:** Build a *hardware-software* interface using **multi-threading** techniques to synchronize object detection results with digital inputs on an **Omron PLC** through a relay module.
3. **Performance Evaluation:** Measure system performance based on detection accuracy metrics (mAP), inference latency (ms), and the success rate of physical sorting of geometric objects in *real-time*.

1.4. Activity Benefits

For Students: Enhance competence in **Edge AI optimization**, PLC programming, and developing system architectures that integrate high-level computer vision with industrial hardware control.

For Institutions: Contribute to research output in the fields of **Applied Artificial Intelligence** and Robotics, and serve as a technical reference for future student research regarding AI integration in factory infrastructure.

BAB II

IMPLEMENTATION METHOD

2.1. Scope of Activities

The scope of activities is focused on developing the perception-to-action workflow: starting from the preparation of geometric object datasets (Cube, Cylinder, Prism), training the YOLOv8 model, configuring hardware acceleration via PCIe M.2 HAT+, to setting up logic on the Omron CP1H PLC to drive a **5-axis robot arm.**

2.2. Implementation Method

1. Phase 1: Model Development: Data acquisition, annotation, and YOLOv8 training, followed by the model quantization process for the Hailo-8 NPU architecture.
2. Phase 2: Hardware Integration: Wiring between Raspberry Pi 5 GPIOs and relay modules to send trigger signals to the Omron PLC inputs.
3. Phase 3: Software Engineering: Python-based program development using *threading* to ensure the AI inference process runs simultaneously with sensor monitoring and relay activation processes.
4. Phase 4: PLC Programming: Designing ladder logic on CX-Programmer to convert signals from the Raspberry Pi into stepper motor movement commands on the robot arm.

2.3. Time and Place

This research internship will be conducted during the semester, spanning approximately 5 months. The research activities will be carried out at the INSYDE Studio of UPN "Veteran" Jawa Timur Computer Science Building 2 which provides the complete 5-Axis Robot with AI and IoT setup, computational facilities, and a supportive academic environment necessary for conducting industrial network experiments.

2.4. Roles and Responsibility of the Student

1. **Conducting a comprehensive literature review** on Edge AI implementation, YOLOv8 architecture optimization for hardware accelerators, and industrial automation protocols between single-board computers and PLCs.
2. **Collecting, preprocessing, and analyzing image datasets** of geometric objects (Cube, Cylinder, Prism) and performing model training and quantization to ensure compatibility with the Hailo-8 AI Accelerator.
3. **Designing and implementing the Integrated Edge AI-PLC Framework**, focusing on the seamless communication between the YOLOv8 inference output on Raspberry Pi 5 and the digital input triggers of the Omron CP1H PLC.

4. **Evaluating system performance** by measuring real-time inference speed (FPS), detection accuracy (mAP), and the reliability of the robotic arm's sorting actions based on the AI-generated triggers.
5. **Preparing research documentation**, compiling technical findings regarding hardware-to-hardware communication latency, and writing the final research report and a scientific paper intended for journal publication.

2.5. Outputs

1. **Functional Prototype:** A high-performance Edge AI-based sorting conveyor system featuring a Raspberry Pi 5 and Hailo-8 AI Accelerator, fully integrated with a PLC-controlled 5-axis industrial robot arm.
2. **Scientific Manuscript:** A research paper detailing the hardware-software integration and performance metrics, prepared for submission to an international or SINTA-ranked journal.
3. **Comprehensive Technical Documentation:** Including electrical wiring diagrams, optimized YOLOv8-Hailo Executable Format (.hef) files, Python source code with multi-threading implementation, and PLC Ladder Logic (CX-Programmer) configurations.

BAB III

CONCLUSION

The proposed project, **"Edge AI-Based Industrial Sorting System using YOLOv8 on Raspberry Pi 5 with Omron PLC Integration,"** serves as a strategic solution to bridge the gap between high-level Artificial Intelligence and deterministic industrial control systems. By leveraging the **Raspberry Pi 5** and the **Hailo-8 AI Accelerator**, this research aims to eliminate the common computational bottlenecks associated with real-time deep learning on edge devices.

Through the integration of **YOLOv8** for high-speed object classification and **PLC Omron** for precise mechanical execution, this system is expected to deliver a low-cost yet high-performance alternative to traditional industrial vision systems. The implementation of **multi-threading** logic further ensures that the transition from visual perception to physical action occurs with minimal latency, meeting the rigorous demands of modern manufacturing environments.

In conclusion, the successful realization of this framework will not only demonstrate the feasibility of advanced Edge AI in industrial automation but also provide a scalable model for future developments in smart factory technologies. It is anticipated that this project will contribute significant technical insights into the optimization of neural networks for hardware accelerators and the synchronization of heterogeneous industrial devices.